

Chiral Central Charge from a Single Bulk Wave Function

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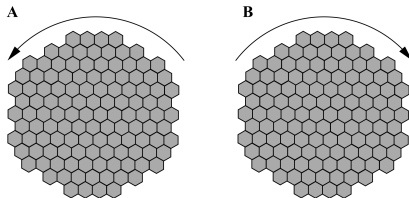
Introduction

Chiral Central Charge: a brief review

- (1+1)D CFT – boundary of a (2+1)D gapped bulk.
- Virasoro algebra

$$\begin{cases} T(z) := T_{00} - T_{11} + 2iT_{01} = \sum_{n \in \mathbb{Z}} \frac{1}{z^{n+2}} L_n \\ \bar{T}(\bar{z}) := T_{00} - T_{11} - 2iT_{01} = \sum_{n \in \mathbb{Z}} \frac{1}{\bar{z}^{n+2}} \bar{L}_n \end{cases} \quad \begin{cases} [L_n, L_m] = (n-m)L_{n+m} + \frac{1}{12} c(n^3 - n)\delta_{n,-m} \\ [\bar{L}_n, \bar{L}_m] = (n-m)\bar{L}_{n+m} + \frac{1}{12} \bar{c}(n^3 - n)\delta_{n,-m} \\ [L_n, \bar{L}_m] = 0 \end{cases}$$

- Chiral central charge (CCC): $c_- := c - \bar{c}$.
- Edge current: $I = \frac{\pi}{12} c_- T^2$



- A. Cappelli et al., Nucl. Phys. B **636**, 568 (2002)
- A. Kitaev, Ann. Phys. **321**, 2 (2006)

Topological ordered states:

- Topological entanglement entropy
- Entanglement spectrum
- Topological S and T matrices (mutual- and self-statistics)

$$S_{ab} = \frac{1}{\mathcal{D}} \text{a} \text{b}$$

$$T_{ab} = \theta_a \delta_{ab}$$

$$= \frac{1}{d_a} \text{a} \text{a} \delta_{ab}$$

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$$T_{ab} = \theta_a \delta_{ab} \\ = \frac{1}{d_a} \text{a} \text{a} \delta_{ab}$$

Missing piece: Chiral Central Charge (CCC)

- Gapless edge excitations
- TRS breaking
- Bulk-boundary correspondence

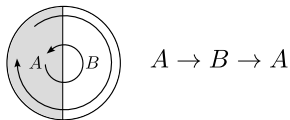
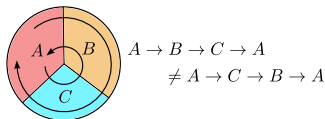
$$\underbrace{\frac{1}{\mathcal{D}} \sum_a d_a^2 \theta_a}_{\text{bulk}} = \underbrace{e^{2\pi i c_- / 8}}_{\text{boundary}}$$

Question: A succinct closed-form formula that relates CCC to the ground state entanglement of a single wave function is missing.

Motivation

Strategy: devise a quantity with

- TRS odd
- Loop nature (chiral) $\Rightarrow$ Tripartition
- Entanglement Hamiltonian
- Topological invariant
- Energy current $I_{v \rightarrow u} = i \langle [h_u, h_v] \rangle$



Definition (Modular commutator)

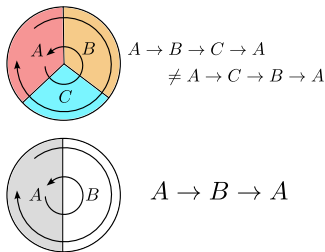
$$J(A, B, C) := i \text{Tr} (\rho_{ABC} [K_{AB}, K_{BC}]) = i \langle [K_{AB}, K_{BC}] \rangle \in \mathbb{R}$$

$K_X \equiv -\ln \rho_X$ is the entanglement Hamiltonian of the subsystem X .

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$K_X \equiv -\ln \rho_X$ is the entanglement Hamiltonian of the subsystem X .

$$J(A, B, C)_{\rho^*} = i\text{Tr}(\rho_{ABC}^*[K_{AB}^*, K_{BC}^*]) = i(-iJ(A, B, C)_{\rho})^* = -J(A, B, C)_{\rho}$$

Methods and Results

Quantum Markov Chain

Definition (conditional mutual information)

$$I(A : C|B) := S_{AB} + S_{BC} - S_B - S_{ABC}$$

Recall that $I(A : C) := S_A + S_C - S_{AC}$, we have

$$\begin{aligned} I(A : C|B) &= I(A : BC) - I(A : B) = I(C : AB) - I(C : B) \\ &= I(AB : BC) - S_B \end{aligned}$$

A **quantum Markov chain** is a state with the tripartition A, B, C with $I(A : C|B) = 0$.

Theorem (D. Petz (2003))

$$I(A : C|B) = 0 \Leftrightarrow K_{ABC} = K_{AB} + K_{BC} - K_B$$

• D. Petz, Rev. Math. Phys. **15**, 79 (2003)

This theorem will be used to decompose the entanglement Hamiltonians.

Quantum Markov Chain

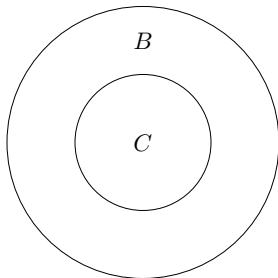
An important fact is that *the modular commutator vanishes if the state ρ together with the tripartition A, B, C is a quantum Markov chain.*

Because

$$I(A : C|B) = 0 \Leftrightarrow K_{ABC} = K_{AB} + K_{BC} - K_B$$

then, if the state is a quantum Markov chain, then

$$\begin{aligned} J(A, C, B) &= i\text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) \\ &= i\text{Tr}(\rho_{ABC}[K_{AB}, (K_{ABC} - K_{AB} + K_B)]) \\ &= i\text{Tr}(\rho_{ABC}[K_{AB}, K_{ABC}]) + i\text{Tr}(\rho_{AB}[K_{AB}, K_B]) \\ &= 0 \end{aligned}$$



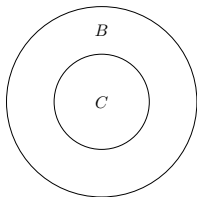
Axiom (A0)

For any state σ , for any configuration of subsystems B (boundary) together with C (core) topologically equivalent to concentric circles,

$$S(\sigma_{BC}) = S(\sigma_B) - S(\sigma_C)$$

Area Law

Area law: $S = \alpha l - \gamma$



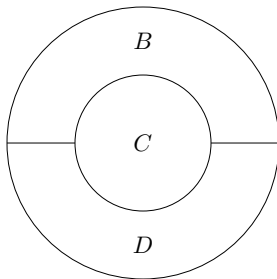
$$\begin{cases} S_B = \alpha(l_1 + l_2) - 2\gamma \\ S_C = \alpha l_1 - \gamma \\ S_B = \alpha l_2 - \gamma \end{cases} \Rightarrow S_{BC} + S_C = S_B$$

Let E be the environment that purifies the state,

$$I(E : C) \equiv S_E + S_C - S_{EC} = S_{BC} + S_C - S_B = 0$$

For any subspace $A \subset E$ of the environment, the subadditivity implies,

$$I(A : C) \leq I(E : C) = 0$$

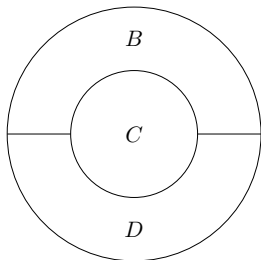


Axiom (A1)

For any state σ , for any configuration of subsystems B, D (two subsystems that bipartite the boundary) together with C (core), topologically equivalent to the above figure,

$$S(\sigma_{BC}) + S(\sigma_{CD}) - S(\sigma_B) - S(\sigma_D) = 0$$

Quantum Markov Chain from Area Law



From Axiom **A1**, we have

$$\begin{aligned} & S_{BC} + S_{CD} - S_B - S_D \\ &= S_{BC} + S_{EB} - S_B - S_{EBC} \\ &= I(E : C|B) = 0 \end{aligned}$$

Where E is the purifying environment.

Let $A \subset E$ be a space outside the system, according to the *subadditivity of the entropy*, we have

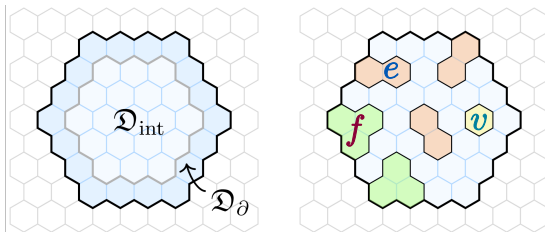
$$I(A : C|B) \leq I(E : C|B) = 0$$

That is to say

$$\text{Axiom A1} \Rightarrow I(A : C|B) = 0 \xrightarrow{\text{Petz}} K_{ABC} = K_{AB} + K_{BC} - K_B$$

- **Area law** (Axiom **A1**) implies **quantum Markov chain**.
- The Entanglement Hamiltonian can be decomposed into smaller supports.

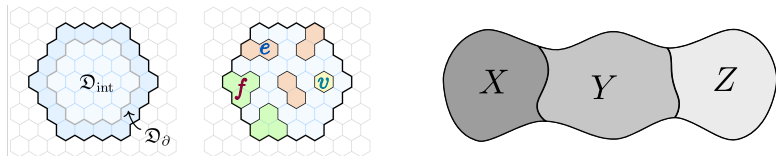
Decomposition of Entanglement Hamiltonian



$K_{\mathcal{D}}$ is the entanglement Hamiltonian of a disk $\mathcal{D}$.

- Coarse-grain the system into a triangular lattice of supersites.
- The area law allows us to decompose the entanglement Hamiltonian using the quantum Markov chain trick.
- Three types, f (face), e (edge), and v (vertex), of subsystems cannot be separated further by the quantum Markov chain trick.

Decomposition of Entanglement Hamiltonian



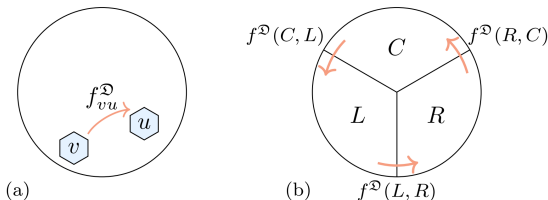
Decomposition procedure

- Do coarse-graining to define elementary subsystems.
- X and Z are disconnected, but both of them are connected to Y .
- Area law axiom implies $I(X : Z|Y) = 0$, thus $K_{XYZ} = K_{XY} + K_{YZ} - K_Y$.
- Continue the process until only f , e and v remain.

Eventually, we will have:

$$K_{\mathcal{D}} = \sum_{f \in F(\mathcal{D})} K_f - \sum_{e \in E(\mathcal{D}) \setminus E(\mathcal{D}_{\partial})} K_e + \sum_{v \in \mathcal{D}_{\text{int}}} K_v$$

Energy Current and Modular Current



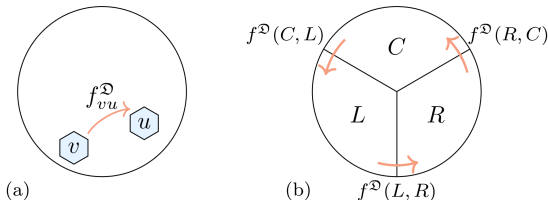
Suppose the Hamiltonian describing the system is local, $H = \sum_u h_u$

$$\frac{\partial}{\partial t} \langle h_u \rangle = -i \langle [H, h_u] \rangle = \sum_v i \langle [h_u, h_v] \rangle \equiv \sum_v I_{v \rightarrow u} = - \oint_u \mathbf{J} \cdot d\mathbf{S}$$

then we can follow this to define the energy current from u to v as

$$I_{v \rightarrow u} = i \langle [h_u, h_v] \rangle$$

Energy Current and Modular Current



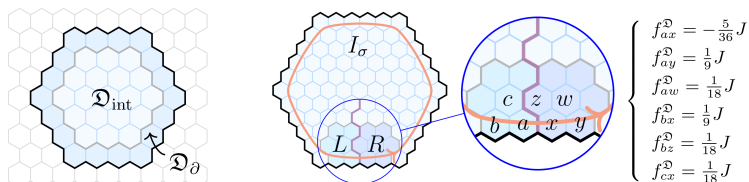
Therefore, the modular current is defined in the same way,

$$f_{v \rightarrow u}^{\mathfrak{D}} := i \left\langle \left[\tilde{K}_v^{\mathfrak{D}}, \tilde{K}_u^{\mathfrak{D}} \right] \right\rangle, \quad f^{\mathfrak{D}}(L, R) := \sum_{v \in L} \sum_{u \in R} f_{v \rightarrow u}^{\mathfrak{D}}$$

where $\tilde{K}_v^{\mathfrak{D}}$ collects all terms whose support contains site v ,

$$K_{\mathfrak{D}} = \sum_{f \in F(\mathfrak{D})} K_f - \sum_{e \in E(\mathfrak{D}) \setminus E(\mathfrak{D}_{\partial})} K_e + \sum_{v \in \mathfrak{D}_{\text{int}}} K_e \Rightarrow \sum_{v \in \mathfrak{D}} \tilde{K}_v^{\mathfrak{D}}$$

Energy Current and Modular Current



- $f_{u \rightarrow v}^{\mathcal{D}} = 0, \quad \forall v, u \in \mathcal{D}_{\text{int}}$. That is, there is only **edge current**.
- Every modular commutator can be reduced to $J \equiv J \left(\begin{array}{c} v \\ w \quad u \end{array} \right)$.
- The edge current can be calculated: $I_{\sigma} \equiv \sum_{u \in L} \sum_{v \in R} f_{u \rightarrow v}^{\mathcal{D}} = \frac{1}{4}J$
- The edge modular current should have the same form as the energy current, $I_{\sigma} = I(T=1) = \frac{\pi}{12}c_- T^2 \Big|_{T=1}$
- $J = \frac{\pi}{3}c_-$, a formula relates CCC to the ground state entanglement.

Conclusions:

- The **modular commutator** can be treated as a link that bridges the **many-body entanglement** with the **chiral central charge (CCC)**.
- For gapped systems, the area law is suggested to hold, which gives rise to the **quantum Markov chain** feature of the system.
- Entanglement Hamiltonians can be decomposed into small pieces via the quantum Markov chain trick, which **makes the modular commutator topological**.
- It is conjectured that two gapped Hamiltonians having the same bulk reduced density matrix generate the same edge current $I = \frac{\pi}{12}c_-T^2$.
- A succinct formula relating the CCC and elementary modular commutator is derived: $J = \frac{\pi}{3}c_-$.

Problems:

- The area law axioms are not guaranteed to hold.
- Violation of area law